Grant for Surgical Robotic Arm for Tympanoplasty Surgical Procedure: Repairing Holes in the Eardrum Through Bio Graft Placement Through the Ear Canal

- Uses personalized pre planned surgical route.
- Inserts the graft through the ear canal, the grafts back layer sliding through the perforation, holding it in place.
- Eliminates possible hand tremors.
- Can navigate the delicate ear canal structure.
- Aligns the graft past human hand precision.
- Replaces the need for an invasive surgical route or anesthesia.
- Surgery can be performed with the patient sitting upright in any office.

Tympanic Membrane Perforations

Over 320 million people suffer from Tympanic Membrane Perforations.

Tympanic membrane perforations are holes in the eardrums thin layer of tissue.

Caused by

- Infections
- Air pressure changes
- Trama
- Myringotomy procedure

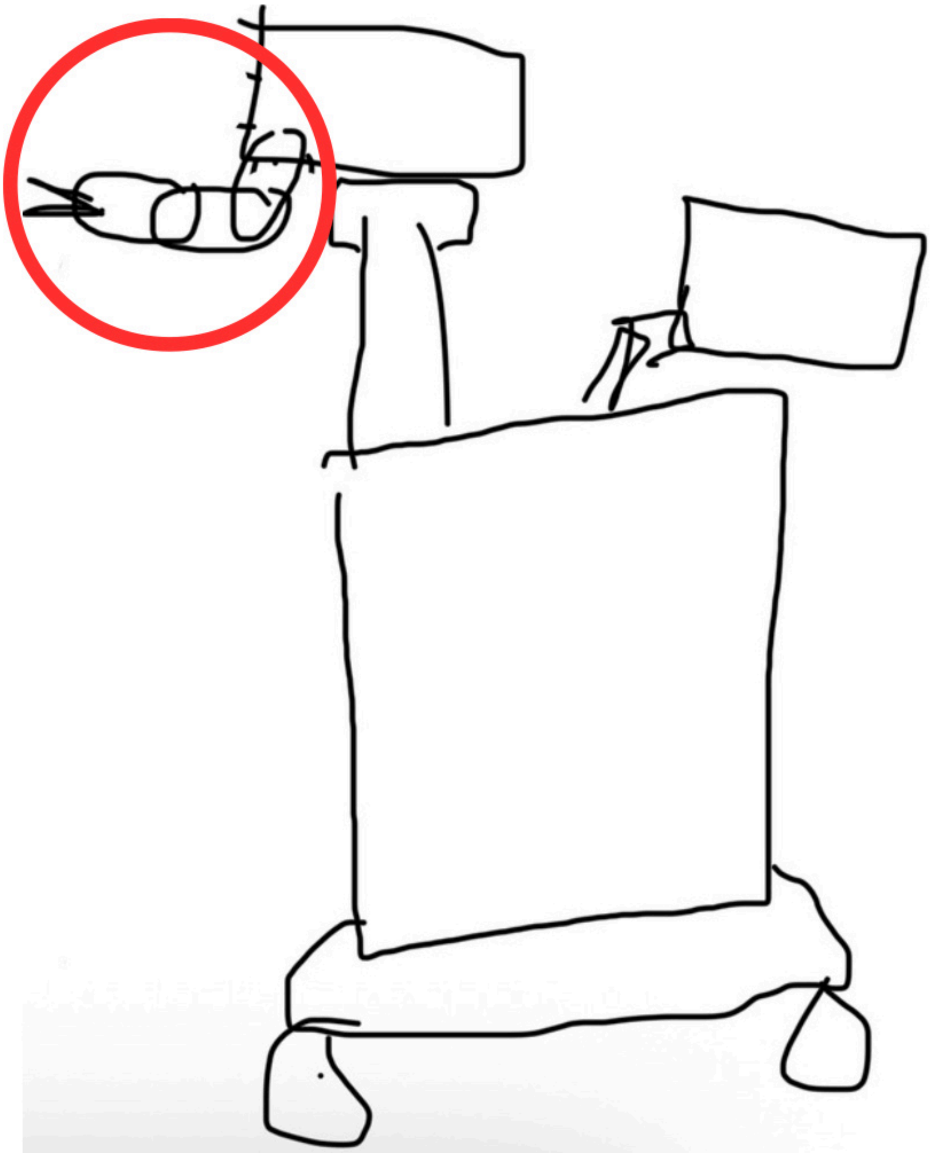
Cause

- Tinnitus
- Vertigo
- Hearing loss
- Ear Infections

Current Surgical Procedure

- Current surgical procedure expensive and inaccessible with low success rates and multiple incisions.
- Entrance to the eardrum from an incision behind the ear
- This incision is painful and takes multiple months to heal
- This access method poses the risk of hitting the hearing bones causing permanent hearing loss.
- Surgery needs anesthesia and takes multiple hours.
- Mistakes can easily occur during surgery, causing punctures of the eardrum.

Layout



The bill of materials below will cover the surgical robotic arm for graft placement and all electrical components. I will print the outer casing for the robotic arm in my robotics workshop at school which provides free filament. I will design the outer casing on solidworks which is also provided by my school. The other parts of the surgical robot including the monitor and cart, I will use my laptop and computer and purchase all other materials needed to build the cart and other accessories.

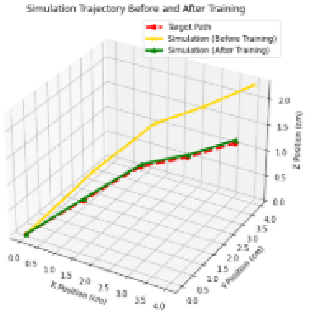
Simulation Training for the Robotic Arm

Below is my previous design for the robotic arm. The new design of the robotic arm will be created and assembled in solidworks.

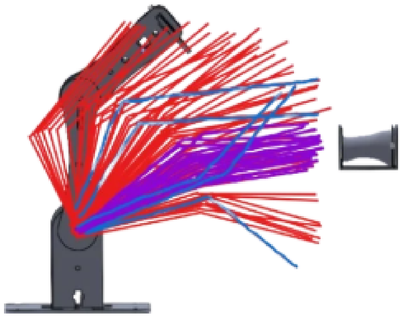
- Assembled in SOLIDWORKS assembly with screws and motors.
- Motion simulation with personalized ear looped for multiple hours.

- The simulation shows where the arm gets stuck, when it touches the canal, and how much each motor must rotate, changing these values and learning from itself.
- The simulation gives XYZ points, joint angles, and sequence of movement used to create a motion path by converting angles to servo commands and putting this information into the code.

Data Analysis



Simulation testing was preformed and each path taken during testing is mapped out to show initial errors, the learning curve, and the best trajectory. After simulation training with the personalized ear, the robot aligned with the ear canal and learned path motion for the surgery.



Previous Design I Want to Advance